Topic 7 — Natural Vegetation & Biodiversity Geography

> **Sources baked in:** NCERT Class 9–11, Champion–Seth vegetation + mangroves + biodiversity (Kullar), **Ghatnachakra** Natural Vegetation (CA-164–170), ISFR/FSI, MoEFCC PA lists, UPPCS Prelims 2018–2025 + older UPPCS/UKPCS vegetation questions **Weight:** ★★★★ — type×rain×region, PA–state, UP Dudhwa/Ramsar, biosphere traps, Project chronology **Last verified:** September 2026 **Current Affairs: Surha Tal (Ballia)** is India's **100th Ramsar** (5 Jun 2026). UP now has **13**. **MISHTI** is the Budget **2023–24** mangrove programme (not 2019).

Current Affairs (this topic)

Year	Fact	Why asked	Source
5 Jun 2026	**Surha Tal / JP Narayan BS (Ballia)** = India's **100th Ramsar**; UP **13th**	Count + latest UP site	MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) / PIB
2025–26	UP Ramsar adds Patna (पटना) BS, Sheikha Jheel, Surha Tal (full names in N.6)	State list (राज्य सूची)	MoEFCC
Budget 2023–24	**MISHTI** mangrove restoration — not a 2019 start	Mangrove scheme year	Budget
Static	Olivia (Coast Guard (तटरक्षक बल), Odisha, Nov–May); coral warming 2024 PYQ	Coastal ops	ICG / PYQ
ISFR	Use the **report year in the question**; 2017 = 21.54% is a frozen PYQ	Figure trap	FSI

Lakes-as-wetlands. This file keeps the **geography** facts.

N.1 Natural Vegetation & Forest Types

Natural vegetation grows without human planting.

- **"Vegetation is the true index of climate"** is linked to **Thornthwaite**, not Köppen's letter codes.
- Controls include climate, soil, relief, biotic factors and time.
- **Species diversity decreases from the equator toward the poles**.
- Tropics / Amazon hold the peak diversity.
- India is a **mega-biodiversity** country.
- Four biodiversity **hotspots** touch India: **Himalaya, Western Ghats, Indo-Burma and Sundaland**.

NCERT five-type frame: Tropical Evergreen (& Semi-Evergreen) | Tropical Deciduous | Tropical Thorn | Montane | Littoral / Swamp (Mangrove).

Type	Rain (cm)	Region	Trees / trait
Tropical evergreen (Selvas)	**>200**	W Ghats, NE, Kerala, WB, Andamans	Multilayer; rosewood, mahogany, ebony, cinchona, rubber, bamboo; **no** complete dry leaf-shed
Semi-evergreen	Slightly less than pure evergreen	W Ghats / NE / Andamans mix zone	Evergreen + deciduous mix
Moist deciduous	**100–200**	Central India, Terai (तराई)–Bhabar (भाबर), E slopes of W Ghats	**Teak, sal**, shisham, neem, mango; **largest area**; sheds in dry season
Dry deciduous	**70–100**	UP, AP, MH, KA, TN; drier peninsula	Rosewood, mango, sandalwood, mahua
Thorn / scrub	**<50–70**	RJ, GJ, SW Punjab–Haryana, rain-shadow / W AP	Babool, ber, khair, khejri, palas, wild date; xerophytes
Montane	altitude	Himalaya; S Indian hills (sholas)	Belts, not plains rainfall ladder; deodar 1500–2500 m west
Littoral / mangrove	tidal	Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र), Godavari (गोदावरी)–Krishna (कृष्णा) deltas + coasts	Sundari; pneumatophores; saline (लवणीय)

- Moist and dry deciduous belong to the same monsoon-forest family, split by rain.
- Evergreen does **not** mean “any green tree in monsoon India.”
- Tea and rubber **estates** are plantations, not climax evergreen.

Champion and Seth (5 × 16 map)

Standard notes split Indian vegetation into **five** heads and **sixteen** sub-types. NCERT’s five climate types usually answer the first question type. The 16 names answer “which is a dry tropical / montane temperate” questions.

Head	Sub-types
Moist tropical	Wet evergreen · Semi-evergreen · Moist deciduous · Littoral and swamp
Dry tropical	Dry evergreen · Dry deciduous · Thorn
Montane sub-tropical	Broad-leaved hill · Moist hill (pine) · Dry evergreen
Montane temperate	Wet temperate · Himalayan moist temperate · Himalayan dry temperate
Alpine	Sub-alpine · Moist alpine scrub · Dry alpine scrub
Share of forest area (class figures)	Type
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About **37%**	Tropical **moist deciduous** (largest)

Head	Sub-types
About **28%**	Tropical **dry deciduous**
About **8%**	Tropical wet evergreen
About **6%**	Sub-tropical moist hill (pine)
About **4%**	Tropical semi-evergreen

- Rain on the **plains** sets the type. Above about **900 m**, **temperature** (altitude) takes over.
- **Soil** is the main control for mangroves and swamps. **Topography** makes alpine and tidal belts local.

Rain table vs NCERT ladder

- Rain above **200 cm** (many notes put wet evergreen above **250**): rain forest.
- Rain of **100–200 cm**: monsoon deciduous (moist).
- Rain of **50–100 cm**: drier deciduous / savanna-like.
- Rain of **25–50 cm**: dry thorn (semi-arid).
- Rain under **25 cm**: desert.
- NCERT thorn is **under 50–70 cm**.
- Keep **moist deciduous** as the **largest** type.
- Do not mix two rain tables in one answer.

Biodiversity

- **Biodiversity** is the variety of life. It is split as **genetic**, **species**, and **ecosystem** (पारिस्थितिकी तंत्र) (ecological) diversity.
- **Species richness** counts how many species are present.
- **Evenness** describes how evenly they are abundant.
- Same richness can hide a few dominant species.
- **Alpha** diversity is diversity **inside** one ecosystem.
- **Beta** diversity is change **between** ecosystems.
- **Gamma** diversity is diversity of a whole **region**.
- Species diversity **falls from the equator toward the poles**.
- Amazon is the peak.
- Insects are the richest animal group.
- Fungi outnumber fish, amphibians, reptiles and mammals combined.
- India holds about **2.4%** of land, about **7–8%** of recorded species, and about **18%** of people — a mega-diverse country.
- Genetic fact: more than **50,000** rice strains and about **1,000** mango varieties.
- An **endemic** species is found only in that place.
- Indigenous is not the same if it also lives elsewhere.
- A **keystone** species is one whose loss shifts the ecosystem (top predators; some exclusive pollinators).

- A **flagship** species is a public mascot (tiger, panda).
- A **foundation** species is a dominant structure-builder (coral, kelp).
- India sits in **two realms**: Himalaya ≈ **Palearctic**; the rest ≈ **Indomalayan / Malayan**.
- Five biomes: tropical humid forest · tropical dry/deciduous · desert/semi-desert · coniferous forest · alpine meadow.
- Wildlife planning uses **10 biogeographic zones** (Trans-Himalaya, Himalaya, Desert, Semi-arid, Western Ghats, Deccan (दक्कन), Gangetic plain, North-east, Islands, Coasts) and about **25 provinces**.

Four hotspots (Conservation International)

- **Himalaya** is **one** hotspot covering the Indian Himalaya. Do **not** count Eastern Himalaya and Western Himalaya as two separate hotspots.
- **Western Ghats and Sri Lanka** is the second.
- **Indo-Burma** covers north-east India and neighbours. **Nicobar is not** in this box.
- **Sundaland** covers **Nicobar** (with Indonesia–Malaysia). It does **not** make the Andaman group the Sundaland fact, and it is **not** a joint "Andaman–Nicobar hotspot."
- Do **not** write the four as Eastern Himalaya, Western Himalaya, Western Ghats and Andaman–Nicobar.
- Hotspots are **not** tropics-only. Extra-tropical examples include the **Mediterranean**, **south-west Australia**, and the **Chilean** temperate / winter-rainfall forest.
- Myers' idea is high **endemism** plus heavy **habitat (वास स्थान) loss**. Do not equate four hotspots with **18** biosphere reserves.

World vegetation matches

These eight rows are two lists. Split them that way.

Same climate, four local names

Winter-rain, summer-dry shrubland is one climate with four regional names.

Name	Where	Memory
Maquis	Mediterranean coast	Maquis = Mediterranean
Chaparral	California	Both start with C
Matorral	Chile	Both have "l"
Fynbos	South Africa (Cape)	Fynbos = fine bush of the Cape

- Do not swap them.
- Maquis is not California.
- Fynbos is not Chile.

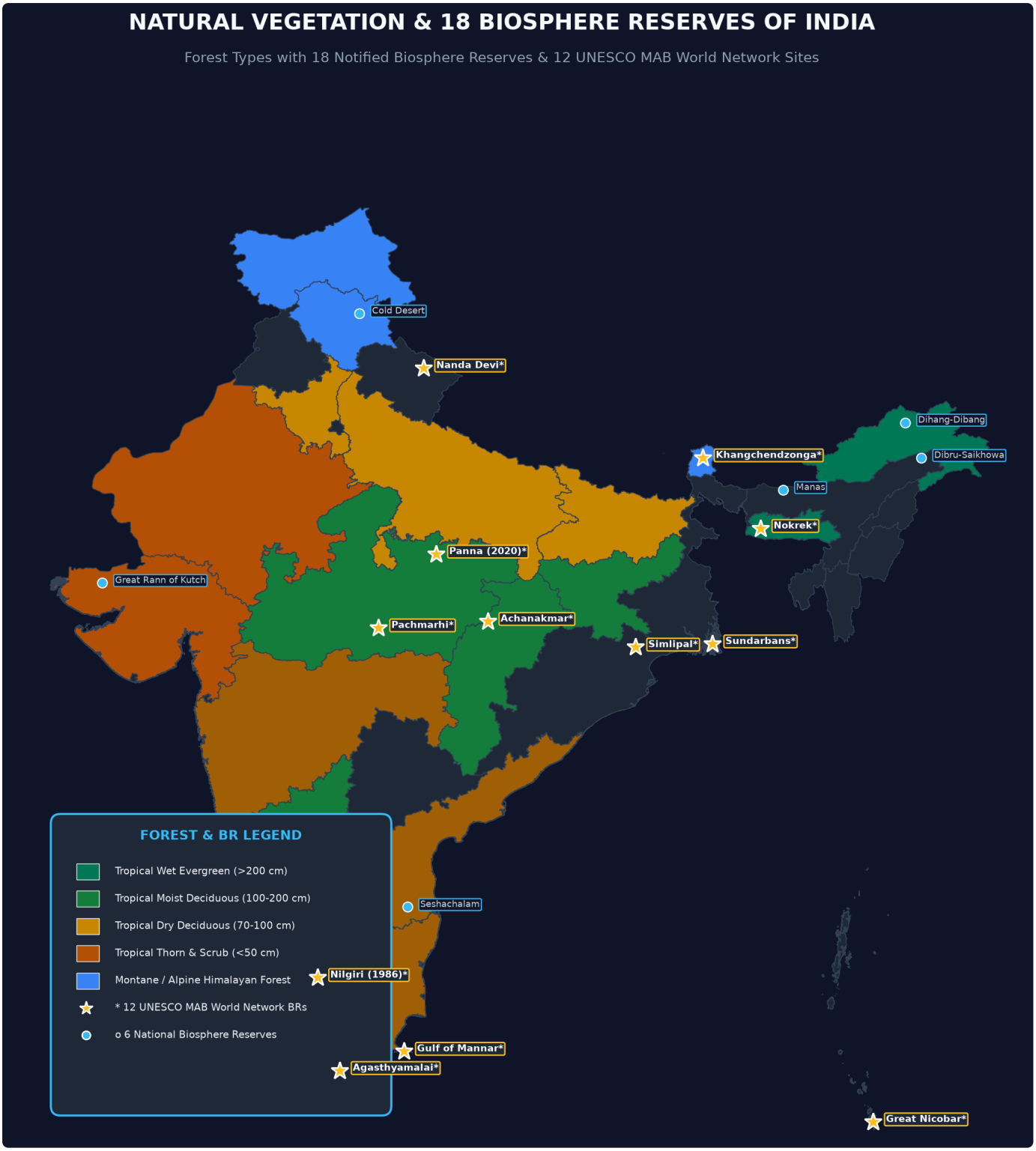
One plant, one belt

Plant	Belt	Why
Epiphytes (orchids on trees)	Equatorial rainforest	Need constant wet heat and a canopy

Plant	Belt	Why
Acacia	Sahara / arid Africa	Thorn tree of dry tracts
Baobab	Savanna	Fat-trunk tree of grassland with scattered trees
Cedars	Mediterranean	Atlas / Lebanon cedar, not equatorial forest

- **Cedar** and **Maquis** can both sit in the Mediterranean.
- Cedar is the **tree**.
- Maquis is the **shrub thicket**.

N.2 Type Cards



Evergreen & semi-evergreen

Tropical evergreen needs equable heat and heavy rain. These forests are also called **Selvas**.

Note	Standard figure	Alternate note you may see
Rain	**>200 cm**	Wet evergreen often written **>250 cm**
Mean annual temperature	Above about **22°C**	Some notes use about **25–27°C**

Note	Standard figure	Alternate note you may see
Humidity	Near **70%**	Some notes write **>77%**
Dry season	Almost none / very short	Short dry season in wet-evergreen notes

- Different species shed leaves at different times, so the canopy stays green year-round.
- Trees are **mesophytes** (not cactus, not water lily).
- Height is about **45–60 m**.
- The canopy is closed. Epiphytes fight for light. Undergrowth holds bamboo, fern, and orchid.
- Hardwood is valuable but **hard to log** because there are no pure stands, climbers are dense, and roads are poor.
- Major trees include mahogany, rosewood, ebony, mesua, white cedar, and jamun.
- Other useful species include **cinchona**, **rubber**, **coconut**, and **bamboo**.

Where it grows

- Windward **Western Ghats** (पश्चिमी घाट) (about **500–1370 m**).
- Assam–Meghalaya–Tripura–Mizoram / **Purvanchal** (पूर्वांचल).
- Wet pockets of **West Bengal**, **Kerala**, and **Andaman–Nicobar**.
- Parts of the western coastal plain.
- This is **not** a Uttar Pradesh plains type.

Star facts

- **Cinchona** grows in Assam, Kerala, and West Bengal evergreen belts.
- Cinchona does **not** grow as a Chhattisgarh dry-deciduous fact.
- **Silent Valley (Kerala)** is the classic tropical evergreen / rain-forest pair.
- **Mahogany** belongs here (tropical evergreen), **not** in the Himalayan set.

Semi-evergreen (the transition)

- Rain is about **200–250 cm**, with a longer dry season than true rain forest.
- Trunks are often **buttressed**.
- Stands are more **gregarious** (purer), so timber is easier than in true rain forest.
- Belts include the western coast, Assam, lower eastern Himalaya, Odisha, and the Andamans.
- Trees include laurel, rosewood, champa (चम्पा), mango, and Indian chestnut.

Deciduous (monsoon forest)

India’s workhorse timber and wildlife forest. Also called **monsoon forests**. Trees shed leaves in the dry season to conserve water.

Type	Rain (usual)	Area	Signature trees
Moist deciduous	**100–200 cm**	Largest share (~**37%**)	**Teak**, **sal**, shisham, neem, mango

Type	Rain (usual)	Area	Signature trees
Dry deciduous	**70–100 cm** (some notes widen to 100–150)	UP, AP, MH, KA, TN; drier peninsula	Rosewood, mango, sandalwood, mahua
Whole deciduous family (wide notes)	Sometimes written **70–200 cm**	Same monsoon-forest family	Split by rain, not by a different “family”

Moist deciduous

- Temperature is about **27°C**. Humidity is about **60–75%**.
- Leaves drop in **spring–early summer** (April–May can look bare).
- Irregular top storey is about **25–60 m**, with buttresses and undergrowth.
- **Teak** and **sal** form useful **pure stands**, so these forests are easier to exploit than evergreen.
- Belt includes eastern slopes of the Western Ghats, the north-eastern peninsula, **Bhabar and Terai**, Chota Nagpur, eastern MP–Chhattisgarh, Odisha, parts of West Bengal, Manipur–Mizoram, and the Andamans.
- Hardwoods dominate furniture timber.
- Madhya Pradesh leads among major states for dense deciduous / teak area.
- Some older notes also list teak and sal under dry deciduous, but the standard textbook association is **moist deciduous**.

Dry deciduous

- This is the drier twin of moist deciduous.
- Found widely in **Uttar Pradesh** (उत्तर प्रदेश), Andhra Pradesh, Maharashtra, Karnataka, and Tamil Nadu, and from the central Gangetic plain toward southern India.
- Wetter side grades into moist deciduous. Drier side grades into thorn.
- Canopy is closed but uneven, with grass on the floor.

Clearance and Uttar Pradesh

- Large tracts of both deciduous types have been **cleared for crops**.
- Fire and grazing thin what remains.
- UP’s north belt is Terai teak–sal.
- UP’s south belt is Vindhyan Sonbhadra (सोनभद्र)–Mirzapur (मिर्ज़ापुर)–Chandauli.
- **Butea monosperma** (**Palash / Dhak**) is called the **flame of the forest**.
- It is the **state flower of Uttar Pradesh**.
- It also appears in thorn / dry belts as **palas**.

Tropical dry evergreen

- A **strange evergreen** on the **Tamil Nadu / Coromandel** coast with only about **100 cm** rain, mostly from the **north-east monsoon** (Oct–Dec).
- Short trees (to about **12 m**), closed canopy; jamun, tamarind, neem. Bamboo and grass are not conspicuous.
- Most of it has been cleared for farms or **casuarina** (N-fixing coastal windbreak and fuelwood).

- Do **not** confuse with Western Ghat wet evergreen.

Thorn / scrub (arid / tropical thorn)

Also called **tropical thorn forests** or arid forests. Plants are **xerophytes**.

Note	Figure / trait
Rain	Often **<75 cm** (NCERT / many notes **<70 cm**)
Humidity	**<50%**
Temperature	About **25–30°C**
Tree height	Only about **6–10 m**, scattered
Dominants	**Acacia** and **Euphorbia**

- The dry season is long.
- Spines, small leaves, deep roots, and thick bark help plants survive drought (सूखा).
- Plants may stay leafless for much of the year.
- Grasses and shrubs are common.

Where it grows

- South-west Punjab and south-west Haryana.
- Rajasthan and Gujarat (Kachchh–Saurashtra).
- Rain-shadow of the Western Ghats, including western Andhra Pradesh.
- Classic look: open stunted forests with long roots and sharp thorns.

Species & product facts

- Important species include babool, ber, wild date palm, khair, neem, khejri, and palas.
- **Katha** is obtained from **khair** wood.
- Khair also appears in sub-tropical Shiwalik (शिवालिक) / western Himalayan scrub questions (UKPCS).
- **Khejri** is the multipurpose social-forestry tree of arid India (“king of the desert”).
- Khejri gives fuel, charcoal, fodder, green manure, and goat-proof fences.
- Khejri is frost-resistant and drought-resistant.
- Westward, this belt grades into Thar (थार) desert vegetation.
- It is **not** “desert with zero plants.”

Montane (altitude elevator)

Montane forests are an **altitude elevator**, not a plains rain ladder. Above about **900 m**, temperature (height) rules the belt. Mountain forests split into two sets: **northern (Himalayan)** and **southern (peninsular hills)**.

Quick Himalayan stack (foothills up to snow)

Height (approx.)	Belt	What you see
Foothills	Tropical / sub-tropical deciduous	Warm lower slopes
About **1000–2000 m**	Wet temperate oak–chestnut	Broad-leaf temperate start
About **2000–3000 m** (southern slopes)	Chir / deodar / pine / spruce / fir	Conifer timber belt
Higher	Alpine juniper / rhododendron / birch	Scrub toward snowline
Highest usable pasture	**Bugyals** (alpine meadows)	Then permanent snow

Champion–Seth montane cards (read one belt at a time)

1. Sub-tropical broad-leaved hill forests

- Height is about **1000–2000 m**.
- Main belt is the **eastern Himalaya east of 88°E**.
- Trees are evergreen **oak**, **chestnut**, and **ash**.
- Stunted versions also sit on Nilgiri–Palani, Mahabaleshwar, Satpura (सतपुड़ा)–Maikal, Bastar, and **Mt Abu**.

2. Sub-tropical moist pine forests

- Height is again about **1000–2000 m**.
- Main belt is the **western Himalaya between 73°E and 88°E**.
- **Chir / chil** grows in fairly pure stands.
- Uses are timber, **resin**, and **turpentine**.
- The same pine story also appears in the Khasi (खासी)–Naga (नागा)–Manipur hills.

3. Sub-tropical dry evergreen

- Height is up to about **1000 m**.
- Belt is **Bhabar** (भाबर), the **Shiwaliks**, and the western Himalaya foothills.
- Cover is low scrub: olive, *Acacia modesta*, pistacia, and the khair–keekar set.

4. Montane wet temperate

- Height is about **1800–3000 m**.
- Rain is about **150–300 cm**.
- Places are the higher hills of **Kerala–Tamil Nadu** and the **eastern Himalaya**.
- Look is mossy evergreen with short trees.

5. Himalayan moist temperate

- Height is about **1500–3300 m**.
- Rain is about **150–250 cm**.
- Spread is **Kashmir to Sikkim**.
- Conifers reach **30–50 m** in fairly pure stands.

- Main conifers are pine, cedar, silver fir, and spruce.
- Oak and rhododendron sit below the tall conifers.
- Timber use includes sleepers and construction wood.

6. Deodar (*Cedrus deodara*) — star temperate fact

- Deodar is a highly valued **western Himalayan** endemic.
- It is abundant at about **1500–2500 m** in the temperate zone.
- Among common options (deodar, pine/chir, sal, teak), deodar sits at the **highest** elevation.
- Memory line: **chir is lower**. **Deodar is higher**.

7. Himalayan dry temperate

- Belt is the **inner ranges** where the south-west monsoon is weak.
- Places include Ladakh (लद्दाख), Lahaul, Kinnaur, Chamba, inner Garhwal (गढ़वाल), and north Sikkim.
- Precipitation is often **snow**, usually **under 100 cm**.
- Trees include deodar, oak, and olive.

8. Alpine belts

- Many notes put alpine from about **2900–3500 m+**; some stretch it to **3000–4900 m**.
- **Sub-alpine** is a mix of conifer and broad-leaf.
- **Moist alpine scrub** holds rhododendron and **birch** up toward the snowline.
- **Dry alpine scrub** above about **3500 m** holds juniper, honeysuckle, and artemisia.
- The **birch** is a Himalayan tree.
- Birch's white paper-like bark was used for writing scriptures.
- Do **not** place birch in the Nilgiris, Aravallis, or Vindhyas.

East vs West Himalaya (high-yield)

Point	Eastern / Central	Western / north-western
Tree-line	About **3600–3800 m**	About **3300–3600 m** (**lower**)
Alpine climb	Climbs **higher**	Stops earlier
Why	Nearer the **equator** and the **sea**; wetter	Drier monsoon side
Resource note	Wet forests	Forests are a **major land resource** of the western Himalayan region

Southern mountain forests (peninsular hills)

- Belts sit on the **Western Ghats**, the **Vindhyas**, and the **Nilgiris**.
- Lower belts are **sub-tropical**.
- Higher belts are **temperate**.
- Temperate patches called **sholas** sit in the **Nilgiris**, **Anaimalai**, and **Palani** hills.

Aspect (north vs south slope)

- Aspect shifts the tree line.
- **Southern** Himalayan slopes get **more rainfall and sunlight**.
- Vegetation is therefore **denser on southern slopes** than on northern slopes.
- Assertion “northern slopes are denser” is **false**.
- Reason “southern slopes get more rain” is **true**.

Species traps for this card

- **Mahogany** is **not** essentially a Himalayan species.
- Himalayan set includes **juniper**, **silver fir**, **spruce**, **deodar**, and **birch**.
- Mahogany belongs with tropical evergreen belts (American origin / Indian evergreen tracts).

Mangrove / tidal (littoral & swamp)

Salt-tolerant **halophytes**. They grow in fresh **or brackish** water. Also called **deltaic / swampy / littoral** forests.

- **Pneumatophores** and **prop roots** slow water, trap silt, and buffer storms.
- Stands grow along coasts in salt (लवणाध्यक्ष) marshes, tidal creeks, mudflats, and estuaries.
- Deltaic stands develop in the deltas of the **Ganga**, **Brahmaputra** (ब्रह्मपुत्र), **Godavari** (गोदावरी), **Krishna** (कृष्णा), Mahanadi (महानदी), and Cauvery.

Core match facts

- The **Sundarbans** (सुंदरबन) (*sundari* / *Heritiera*) form the largest single tidal mangrove block.
- The Sundarbans are Royal Bengal Tiger and crocodile country.
- The Sundarbans are **littoral / tidal** forest — not dry deciduous, not alpine.
- **Bhitarkanika (Odisha)** is the usual **second** Indian mangrove fact.

Mangrove pocket	State
Achra–Ratnagiri	**Maharashtra**
Coondapur	**Karnataka**
Pichavaram	**Tamil Nadu**
Vembanad	**Kerala** (not Karnataka)

- Andaman–Nicobar mangroves are dense and less disturbed.
- West coast mangroves (MH–Goa (गोवा)–KA) are often **scrubby and degraded**.
- Gulf of Kachchh / Kori stands are often **dwarf**.
- Swamp forests hug the Ganga, Mahanadi, Godavari, Krishna, and Cauvery deltas.
- Old uses were fuelwood and boat timber.
- NCERT order-of-magnitude: India holds a notable share of world mangroves (~7% in older NCERT figure).

- **MISHTI** (Mangrove Initiative for Shoreline Habitats and Tangible Income) is the Budget **2023–24** restoration programme.
- It was **not initiated in 2019**.

Species & product facts (Ghatnachakra)

Fact	Note
Flame of the forest	**Butea monosperma** = Dhak / Palash = UP state flower
Katha	From **khair** wood
Khejri	Multipurpose arid / social-forestry tree
Birch	Himalayas; writing-bark tradition
Fern	Vascular, spore-bearing; **no flowers / no seeds**
Oryx vs Chiru	Oryx = hot arid; Chiru = cold high-mountain steppe (स्टेपी) / semi-desert
Teak–Central India	Major teak expansion in MP–CG–Odisha; MP often leads dense deciduous area
Cinchona	Evergreen / Terai–NE match; **not** Chhattisgarh

PYQ — UPPCS Prelims 2018, Q25

'Vegetation is the true index of climate'. This statement is associated with

A. Thornthwaite

B. Koppen

C. Trewartha

D. Stamp

Show answer

Ans: A — Thornthwaite, not Köppen's letter codes.

N.3 Forest Distribution & ISFR

Forest Survey of India (FSI), Dehradun (under MoEFCC), publishes the biennial **India State of Forest Report (ISFR)**.

Core Definitions

- **Forest Cover (वन आवरण):** Satellite canopy mapping. All lands with tree canopy density ≥ **10%** and area ≥ **1 hectare**, **irrespective of ownership or legal status** (includes private orchards, tea estates, and plantations).
- *Very Dense Forest (VDF):* Canopy **> 70%**
- *Moderately Dense Forest (MDF):* Canopy **40%–70%**

- *Open Forest (OF):* Canopy **10%–40%**
- *Scrub:* Canopy **below 10%** (NOT counted in Forest Cover).
- **Tree Cover (वृक्ष आवरण):** Small tree patches, isolated trees, and linear roadside/canal plantings **below 1 hectare** occurring outside recorded forests (Trees Outside Forest / TOF).
- **Recorded Forest Area (RFA) (अभिलिखित वन क्षेत्र):** Lands legally recorded or notified as forest in government revenue/forest records under the Indian Forest Act 1927 (**Reserved**, **Protected**, and **Unclassed**), regardless of whether trees physically grow there.
- **Key Rule:** `Forest Cover ≠ Recorded Forest Area`. Approximately 27% of India's forest cover lies *outside* legally notified RFA.

Active Forest Benchmark (For 2025 & 2026 Exams)

1. All-India Snapshot (Latest FSI Assessment)

Parameter	Latest Figure (For 2025–26 Exams)	% of India's TGA	Trend / Change
Forest Cover	**7,15,343 sq km**	**21.76%**	Net increase across country
Tree Cover	**1,12,014 sq km**	**3.41%**	Trees outside forest (TOF)
Total Green Cover	**8,27,357 sq km**	**25.17%**	Net gain of **+1,445 sq km**

- **Top States by Forest Area: Madhya Pradesh (85,724 sq km)** > Arunachal Pradesh > Chhattisgarh.
- **Top States by Forest %: Mizoram (~84.5%)** > Arunachal Pradesh > Meghalaya (among UTs: Lakshadweep ~90%).
- **Maximum Gainers (Combined Forest + Tree): Chhattisgarh (+684 sq km)** > **Uttar Pradesh (+559 sq km)** & **Odisha (+559 sq km)**.
- **Maximum Forest Cover Gain (Forest only): Mizoram (+242 sq km)** > Gujarat (+180 sq km) > Odisha (+152 sq km).
- **Maximum Loss: Madhya Pradesh (-612.41 sq km)**.

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2. Uttar Pradesh Forest Profile (2025–2026 Data)

Metric	Official Figure	% of UP's Total Area	Notes
Forest Cover	**15,045.80 sq km**	**6.24%**	Satellite canopy ≥ 10%
Tree Cover	**8,950.92 sq km**	**3.72%**	Small patches & roadside trees
Total Green Cover	**23,996.72 sq km**	**9.96%**	UP gained **+559 sq km** (#2 in India!)

- **UP District Extremes:**
- **Highest Forest Cover (Area & %): Sonbhadra (~35.2%)**
- **Lowest Forest Cover (Area & %): Bhadohi / Sant Ravidas Nagar (~0.37%)**
- **Districts with more than 20% Forest Cover:**

- **Chandauli (~21.63%)** and **Shravasti (~20.95%)** have more than 20%.
- **Bahraich is only ~10.15%** (despite Katarniaghat Wildlife Sanctuary, total district cover is under 20%).
- **Near-Nil Forest Land: Jaunpur, Ghazipur, and Ballia** (Purvanchal alluvial plain, officially accepted as having no recorded forest land).
- **The UPPSC 12.8% Trap:** While FSI satellite cover is **6.24%** (9.96% total green cover), UPPSC officially accepted **12.8%** as correct based on state revenue records. Know both figures.

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3. Uttarakhand Forest Profile (UKPCS / 2025–2026 Data)

Metric	Official Figure	% of UK's Total Area	Notes
Forest Cover	**~24,305 sq km**	**45.44%**	Actual satellite canopy mapped
Recorded Forest Area (RFA)	**~38,000 sq km**	**~71.05%**	Legally notified forest in government records
Growing Stock Rank	**Rank #1 in India**	—	Highest increase in forest growing stock nationwide

- **Critical UKPCS Trap (Cover vs. RFA):**
- **71%** is the **legal Recorded Forest Area (RFA)** notified under forest laws.
- **45.44%** is the actual physical **Forest Cover** detected by satellite.
- Do **not** confuse the two in UKPCS prelims options!
- **UK District Extremes:**
- **Largest Forest Area (sq km): Pauri Garhwal** (followed by Uttarkashi and Chamoli).
- **Highest Forest %: Nainital (~71.6%)** (followed by Champawat and Pauri Garhwal).
- **Lowest Forest % & Area: Udham Singh Nagar (~16.8%)** and **Haridwar** (the two plains / industrial-agricultural districts).
- **Mountain Benchmark:** National Forest Policy 1988 mandates a **66% (two-thirds)** forest cover goal for Himalayan hill states.

Policies & Administrative Classification

- **National Forest Policy 1988:** Sets a national target of **33%** total forest and tree cover for India.
- Plains target: 20–25%.
- Hills and mountainous regions target: **66% (two-thirds)** to prevent soil erosion and landslides.
- **Legal Administrative Classes:**
- **Reserved Forests:** Highest degree of protection; all activities (grazing, timber harvesting) prohibited unless specifically permitted.
- **Protected Forests:** Activities permitted unless specifically prohibited.
- **Unclassed Forests:** Other unclassified government or private/community woodlands.

- *(Note: These are legal governance categories under IFA 1927, not NCERT ecological/climatic zones like evergreen or deciduous).*
- **Social Forestry:** Term coined by the National Commission on Agriculture (NCA) in 1976. Refers to raising plantations with community participation outside traditional dense forest reserves (farm forestry, community forestry, agro-forestry, and urban forestry).

N.4 Plant Acclimatization & Adaptations

Acclimatization is an organism adjusting its physiology to a new climate over days or seasons. **Adaptation** is a longer evolutionary trait that helps a species survive in a habitat. Teaching uses these **adaptation types** for vegetation questions.

These names are two lists. Split them that way.

The plant (where it lives)

Type	Hindi	Meaning	Example
Xerophyte	मरुद्भिद	Dry-adapted	Thorn forest: spines, succulence, deep roots
Hydrophyte	जलोद्भिद	Water-adapted	Lotus, hydrilla
Mesophyte	समोद्भिद	Moderate moisture	Most crops / deciduous understorey
Halophyte	लवणमृद्भिद	Salt-adapted	Mangroves
Epiphyte	अधिपादप	Grows on another plant	Equatorial orchids

The mangrove root (not a plant class)

Feature	Hindi	Meaning	Example
Pneumatophore	वात-मूल / श्वसन मूल	Breathing root	Mangrove roots that stick up out of mud

Mangrove is the **plant** (halophyte). Pneumatophore is the **root trick** of that plant.

- A hydrophyte lives **in water** (lotus).
- An epiphyte sits **on a tree** (orchid). It is not a parasite.
- A halophyte lives **in salt** (mangrove).
- A pneumatophore is the mangrove's **breathing root**, not a sixth plant class.

A **xerophyte** lives where water is scarce. It may grow spines instead of broad leaves, store water in fleshy stems, shed leaves in drought, or send roots deep into groundwater. Thorn forests of Rajasthan and Gujarat are classic xerophyte belts.

A **hydrophyte** lives in water or in permanently wet soil. Leaves may be floating or finely dissected. Roots are often weak because water supports the plant. Lotus and hydrilla are standard examples.

A **mesophyte** needs moderate soil moisture. Most crop plants and many deciduous understorey shrubs fall here. They cannot endure long desert drought or permanent flooding.

A **halophyte** tolerates saline soil or brackish water. Mangrove trees are the main Indian example. Many mangroves also grow **pneumatophores**, which are upward breathing roots that take oxygen from air when mud is waterlogged.

An **epiphyte** grows on another plant for support, not as a parasite that steals food. Equatorial orchids on tree trunks are the usual match. Parasites differ because they draw nutrients from the host.

N.5 Protected Areas — Framework

UNESCO MAB and India's Biosphere Reserve Programme

- **UNESCO Man and the Biosphere (MAB)** programme promotes **core + buffer + transition** zoning for conservation with sustainable human use.
- India runs a **national Biosphere Reserve programme** under the **Wildlife (Protection) Act** framework; **18** sites are notified nationally.
- **Nilgiri (1986)** was India's **first** Biosphere Reserve.
- **Core zone** (कोर क्षेत्र) — strict protection, no human extraction.
- **Buffer zone** (बफर क्षेत्र) — limited research and tourism.
- **Transition zone** — settlements, forestry, and agriculture with regulation.
- Several Indian BRs (Nilgiri, Nanda Devi, Sundarbans, Gulf of Mannar, etc.) also sit on UNESCO's **World Network of Biosphere Reserves**.

Category	Fact
National Park	Strictest; no private rights in core idea; habitat + species
Wildlife Sanctuary	Wildlife first; some regulated rights possible
Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र)	**Core** (strict) + **buffer** + **transition** (settlements/use) — UNESCO MAB idea; India list is national programme too
Conservation / Community Reserve (सामुदायिक आरक्षित क्षेत्र)	WLPA later additions — community/landscape gaps
Tiger Reserve	NTCA unit (core+buffer); may contain NP + WLS

- **In-situ** conservation keeps species in their natural habitat (NP, WLS, BR, sacred groves (पवित्र उपवन)).
- **Ex-situ** conservation uses zoos, **botanical gardens**, seed banks and captive breeding.
- 2023: genetic diversity (आनुवंशिक विविधता) in NPs is maintained through **in-situ** conservation.
- Chronology: **WLPA 1972**, then **Project Tiger 1973 (1 Apr)**, then **Project Elephant 1992**, then **BDA 2002**.
- Other projects: Project Rhino, Project Great Indian Bustard.
- India has **18** Biosphere Reserves; **Nilgiri (1986)** is the first.
- Several BRs are on UNESCO's World Network (Nilgiri, Nanda Devi, Sundarbans, Gulf of Mannar, etc.).

High-frequency NP / species

Name	State	Extra
Dachigam	J&K	**Hangul** / Kashmir stag
Keibul Lamjao	Manipur	Only floating NP; **Loktak**; Sangai
Dudhwa	UP	Only UP NP; Lakhimpur Kheri
Simlipal	**Odisha** (ओडिशा)	Not MP
Bandipur (बांदीपुर)	Karnataka	
Manas	Assam	WLS/BR/tiger landscape
Periyar (पेरियार)	Kerala	
Keoladeo	Rajasthan	Birds
Kanha	MP	Tiger
Indravati	Chhattisgarh	2020 match
Mollem	Goa	2020
Kalesar	Haryana	2020
Betla	Jharkhand	2020 (paper OCR “Betva”)

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Biosphere reserves (match gold)

Biosphere	State(s)	Trap
Nilgiri (नीलगिरि)	TN, KL, KA	**First** Indian BR (1986)
Nanda Devi	**Uttarakhand** (उत्तराखंड)	Not UP
Nokrek	**Meghalaya** (मेघालय)	**Not Manipur**
Great Nicobar	A&N	
Gulf of Mannar	Tamil Nadu	Also reefs
Manas	Assam	Correct in 2025
Sundarbans	West Bengal	
Simlipal	Odisha	
Pachmarhi	MP	2021 match
Agasthyamalai	**Kerala and TN**	“Kerala” alone treated correct (2025)
Achanakmar–Amarkantak	MP–CG	

Biosphere	State(s)	Trap
Panna	MP	
Kachchh	Gujarat	
Cold Desert	HP	
Seshachalam	AP	
Khangchendzonga	Sikkim	
Dibru-Saikhowa	Assam	
Dehang-Debang	Arunachal	

PYQ — UPPCS Prelims 2018, Q54

Which is NOT an example of in-situ conservation (स्व-स्थाने संरक्षण)?

A. Biosphere reserve

B. Botanical garden

C. National Park

D. Sacred groves

Show answer

Ans: B — Botanical garden is **ex-situ**.

N.6 Uttar Pradesh — Forests, Parks, Sanctuaries, Ramsar

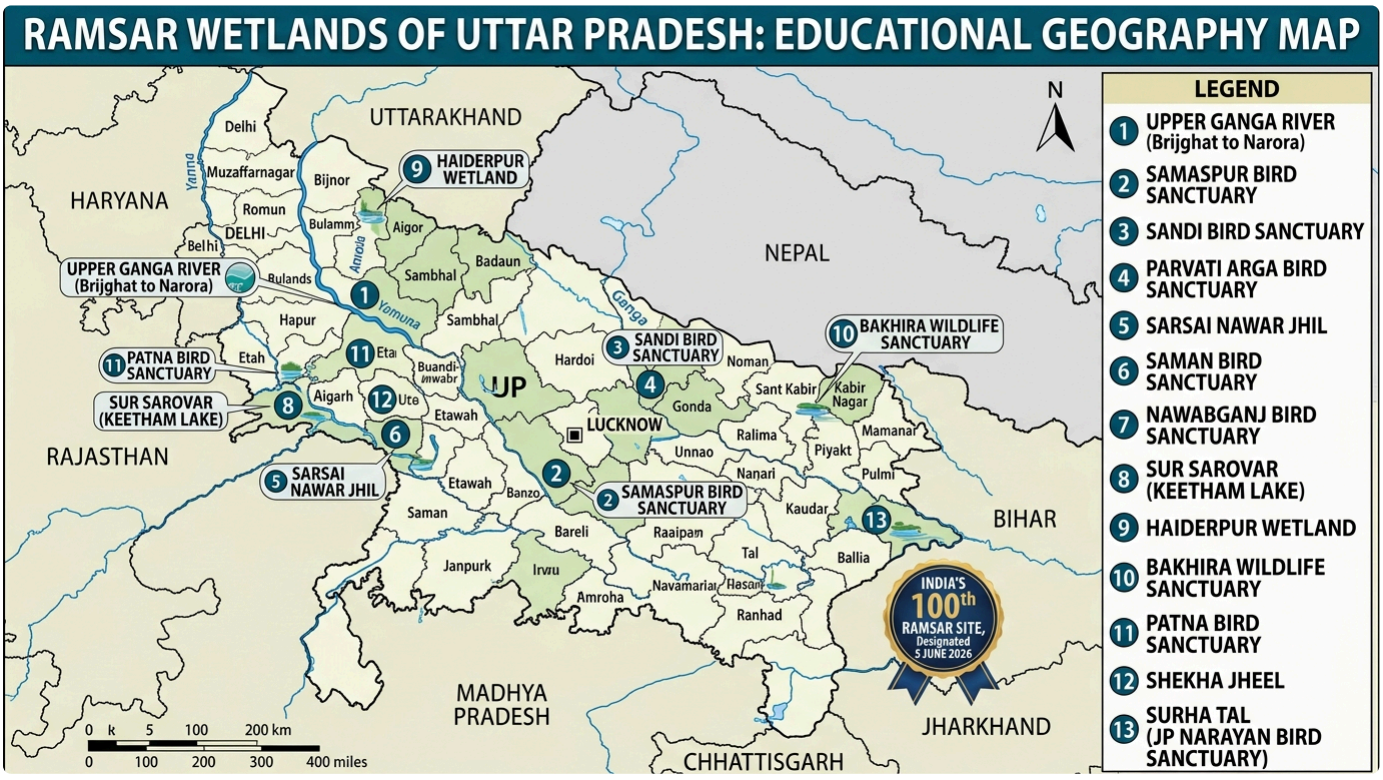
Only one National Park: Dudhwa (Lakhimpur Kheri). Do not invent a second NP in UP.

Name	Type	Belt	Note
Dudhwa	NP	Lakhimpur Kheri	Only UP NP
Dudhwa TR	Tiger Reserve	Kheri + Bahraich complex	**NP + Kishanpur + Katarniaghat**
Pilibhit TR	Tiger Reserve	Pilibhit–Shahjahanpur (शाहजहाँपुर)	Separate Terai TR — not Dudhwa NP
Amangarh TR	Tiger Reserve	**Bijnor**	Corbett buffer/extension
Ranipur TR	Tiger Reserve	**Chitrakoot** (चित्रकूट)	Bundelkhand
Sanctuary	District	Extra	
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Katarniaghat	Bahraich	Est. **1975**	

Name	Type	Belt	Note
Kishanpur	Lakhimpur Kheri	Dudhwa TR	
Chandraprabha	**Chandauli**	not Sonbhadra	
National Chambal (चंबल)	Etawah (इटावा)–Agra	Gharial; **UP–MP–RJ**	
Hastinapur (हस्तिनापुर)	Meerut (मेरठ)–Bijnor–Amroha	Floodplain / swamp deer	
Kaimoor	Sonbhadra–Chandauli	Vindhyan	
Suhelwa	Balrampur–Shravasti–Gonda	Nepal fringe	
Sohagibarwa	Maharajganj	Terai	
Mahavir Swami (स्वामी)	Lalitpur	Bundelkhand	
Ranipur	Chitrakoot	TR base	
Okhla	GB Nagar fringe	Birds	

- Forest belts: Terai-bhabar moist deciduous north; Vindhyan south.
- Sonbhadra often leads absolute (निरपेक्ष) forest area in UP.

UP Ramsar (learn IN vs NOT)



UP total is 13 (June 2026). January 2020 added **six** Ramsar sites to UP.

#	Ramsar site (IN UP)	District / belt
1	Upper Ganga	Brijghat–Narora
2	Nawabganj	Unnao
3	Sarsai Nawar	Etawah
4	Sandi	Hardoi
5	Samaspur	Raebareli
6	Parvati Arga	Gonda
7	Saman	Mainpuri
8	Sur Sarovar (Keetham)	Agra
9	Haiderpur	Muzaffarnagar–Bijnor
10	Bakhira	Sant Kabir (कबीर) Nagar
11	Patna Bird Sanctuary	Etah
12	Sheikha / Shekha Jheel	Aligarh (अलीगढ़)
13	Surha Tal / JP Narayan Bird Sanctuary	Ballia
Name that sounds like UP	Actual state	Trap
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Rudrasagar	**Tripura**	Not UP
Sultanpur	**Haryana**	Not UP
Surinsar–Mansar	**J&K**	Not UP

- **Surha Tal (Ballia)** is India's **100th Ramsar** (designation window 2026; WED **5 Jun 2026** announcement) and UP's **13th**.
- It has been a bird sanctuary since **1991** (later named JP Narayan).
- **Sarsai Nawar** and **Samaspur are** in UP — do not drop them on "NOT in UP" questions.

N.7 Coral, Coastal Ops, Species Programmes

Coral reefs are calcium-carbonate structures built by tiny coral polyps and their symbiotic algae (**zooxanthellae** (ज़ूज़ैंथेली)). They need clear, warm, shallow tropical seas.

Conditions for reef growth

- Mean water temperature is usually about **20–30°C**. Water that is much colder or much hotter stresses the polyps.
- Salinity should stay in a moderate oceanic range. Very fresh river water and extremely high salinity both harm reefs.

- Depth is usually shallow — often within about **45–55 m** — so sunlight reaches the algae for photosynthesis.
- Water must be **clear and low in sediment**. Muddy delta (डेल्टा) mouths clog polyps, so major reefs avoid big river mouths.
- Hard substrate and good circulation help larvae settle and feed.

India's main reef tracts are the **Andaman and Nicobar Islands**, **Lakshadweep** (लक्षद्वीप), the **Gulf of Mannar**, and the **Gulf of Kachchh**.

Coral bleaching happens when heat stress makes corals expel their zooxanthellae. The reef turns white. A bleached coral is not dead at once, but prolonged bleaching can kill it. The 2024 Andaman mass-death framing links bleaching to about a **2°C** rise in Indian Ocean surface water, **not** to cooling.

Coastal operation

- **Operation Olivia** is run by the Indian **Coast Guard**.
- It protects **Olive Ridley** turtles nesting on the **Odisha** coast.
- The main nesting window is about **November to May**, **not** August (the August option is the usual trap).

Species programmes already in this chapter's theme

- **Project Tiger** (प्रोजेक्ट टाइगर) began on **1 April 1973** after the Wildlife Protection Act **1972**.
- **Project Elephant** (प्रोजेक्ट एलीफेंट) began in **1992**.
- Full chronology and reserve framing sit in N.5.
- **MISHTI** supports mangrove restoration and pairs with the littoral / mangrove card in N.2.
